INTRODUCTION

This statistical report aims to identify the most effective classification model for predicting heart disease status, a leading cause of mortality worldwide, using the **heart-disease-dsa.csv** dataset. The analysis focuses on three prominent classification methods: **K-Nearest Neighbours (KNN)**, **Decision Tree (DT)**, and **Logistic Regression (LR)**. Each method is systematically evaluated to determine its predictive accuracy and goodness of fit, with the ultimate goal of identifying the most reliable and effective approach for heart disease prediction.

The **heart-disease-dsa.csv** dataset consists of medical records from 300 individuals, each characterized by a binary response variable, **disease**, indicating the presence or absence of heart disease. Accompanying this response variable are 12 predictor variables, which include both clinical and demographic attributes that are commonly associated with cardiovascular risk. These variables form the basis for model development and the subsequent prediction of heart disease status.

The report begins by defining the response variable and each of the input variables, explaining their relevance to heart disease prediction based on existing research and medical insights. Following this, an **Exploratory Data Analysis (EDA)** is performed on the dataset, utilizing boxplots for numerical input variables and contingency tables for categorical variables. For numerical variables, associations with heart disease status are explored through the median and interquartile range (IQR), identifying differences in the distribution of the disease status. For categorical variables, associations are analysed by calculating and comparing odds ratios, providing insight into the strength of the relationship between each category and heart disease. This analysis serves to identify the most significant predictors of heart disease, which will inform the subsequent modelling phase.

The report then proceeds with the implementation of the classification methods, applying the identified significant features. The optimal version of each method is determined through **5-fold cross-validation**, ensuring the robustness of the models. For both KNN and DT, hyperparameters such as the number of nearest neighbours (**k**) and the minimum split value (**minsplit**) are systematically varied to identify the settings that maximize the True Positive Rate (TPR) or sensitivity. For Logistic Regression (LR), model complexity is addressed by removing features with high p-values, thus simplifying the model. Cross-validation is conducted on this adjusted model, and sensitivity is calculated to assess its performance.

Finally, the report compares the performance of the finalized classifiers—KNN, DT, and LR—based on various performance metrics, including **True Positive Rate (TPR)**, **precision**, **Receiver Operating Characteristic (ROC)** curve, and **Area Under the Curve (AUC)**. These metrics are evaluated on the full dataset, providing a comprehensive assessment of each model's effectiveness. A critical analysis of the advantages and limitations of each classifier follows, considering factors such as model complexity, interpretability, overfitting susceptibility, and computational efficiency. Based on this comparison, the report concludes with a recommendation for the most suitable classifier for predicting heart disease, ensuring that the selected model is both statistically robust and practically applicable in real-world scenarios.

EDA: EXPLORING THE VARIABLES AND ASSOCIATION

Variable	Type	Description	Relevance to Heart Disease Risk
disease	Categorical	1 = Presence, 0 = Absence of heart disease	Direct indicator of heart disease status.
age	Numerical	Age (in years)	Older age increases risk due to accumulated risk factors.
sex	Categorical	1 = Male, 0 = Female	Males at higher risk at younger ages, females' risk increases post-menopause.
chest.pain	Categorical	0 = Typical angina, 1 = Atypical angina, 2 = Non-anginal pain, 3 = Asymptomatic	Angina (0) strongly linked to heart disease, other types less so.
bp	Numerical	Resting blood pressure (mm Hg)	Hypertension damages arteries and increases the risk of heart disease.
chol	Numerical	Serum cholesterol (mg/dl)	High cholesterol generally promotes atherosclerosis.
fbs	Categorical	1 = Fasting blood sugar > 120 mg/dl, 0 = ≤ 120 mg/dl	High blood sugar indicates diabetes, increasing heart disease risk.
rest.ecg	Categorical	0 = Normal, 1 = ST-T wave abnormality, 2 = Left ventricular hypertrophy	Abnormal ECG (1, 2) suggests heart strain, a risk factor for heart disease.
heart.rate	Numerical	Highest heart rate during exercise testing	Lower heart rate indicates poor cardiovascular health, increasing disease risk.
angina	Categorical	1 = Yes, 0 = No (exercise-induced angina)	Angina during exercise signals ischemia, a key heart disease indicator.
st.depression	Numerical	ST depression during exercise ECG (mm)	ST depression indicates ischemia, a strong sign of heart disease.
vessels	Categorical	0–4 = Number of coronary vessels visible by fluoroscopy	More vessels affected indicates more severe heart disease.
blood.disorder	Categorical	1 = Normal, 2 = Fixed defect, 3 = Reversible defect, 0 = Missing	Blood disorders impair circulation or clotting, heightening heart disease risk.

Table 1. General association between input variables and the response variable based on existing medical research.

Variable	Median Difference	IQR Pattern (by Group)	Association with Disease	Usage
Age	Higher in disease group	Wider in no-disease group	Positive (Strong)	Kept
Resting BP	Similar	Slightly more spread in disease group	Weak Positive	Discarded
Cholesterol	Similar	Slightly wider in disease group	None	Discarded
Max Heart Rate	Lower in disease group	Slightly wider + low outliers in disease group	Negative (Strong)	Kept
ST Depression	Higher in disease group	Wider in disease group	Positive (Strong)	Kept

Table 2. Sample association between numerical input variables and the response variable.

Variable	Category	Odds Ratio (vs. Reference)	Interpretation	Usage
Sex	Male (1)	3.61	Strong positive association	Kept
Chest Pain	Atypical	0.08	Strong negative association	Kept
	Non-anginal	0.10	Strong negative association	Kept
	Asymptomatic	0.17	Moderate negative association	Kept
Fasting Blood Sugar	High (1)	1.15	Very weak positive	Discarded
Resting ECG	ST-T abnormal	0.49	Moderate negative	Kept
	LVH	2.51	Strong positive (but small sample size)	Used with caution
Exercise-Induced Angina	Yes (1)	7.41	Very strong positive	Kept
Number of Vessels	1–3	↑ odds (up to 5.67)	Strong positive	Kept
	4	0.71 (low reliability)	Low odds, very small sample	Used with caution
Blood Disorder	Fixed defect	0.14	Strong negative	Kept
	Reversible	1.61	Moderate positive	Kept
	Normal	1.0 (ref)	High (sample size small — interpret carefully)	Used with caution

Table 3. Sample association between categorical input variables and the response variable.

METHODS: BUILDING KNN, DT, and LR CLASSIFIERS

1. CLASSIFIER OPTIMIZATION

To optimize each classifier, hyperparameter tuning using 5-fold cross-validation (CV) is performed to ensure a fair comparison between the methods. The following steps were followed for each classifier:

K-Nearest Neighbours (KNN):

- The number of nearest neighbours (k) was varied from 1 to 50.
- We evaluated each value by using sensitivity or mean TPR (True Positive Rate) of 5-fold cross-validation cycle to find the optimal value.
- Based on the results, the second-best value of k = 3 with the second-highest sensitivity was selected, as it provided the best balance between sensitivity and stability without overfitting. Additionally, the difference in sensitivity of the first and the second-best k values was not significant.

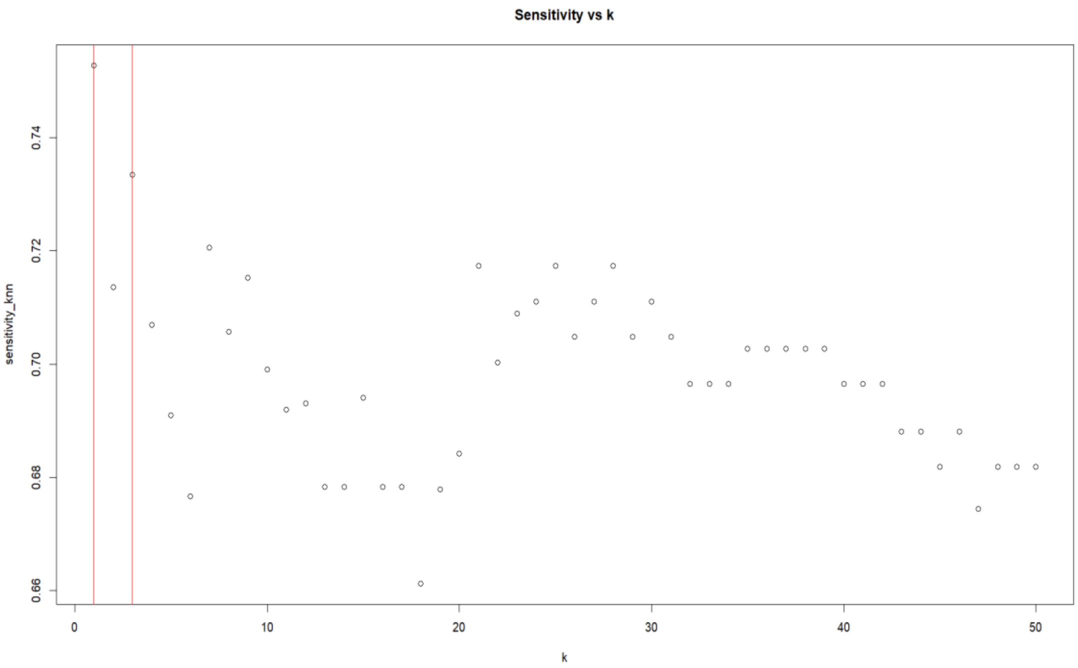

Image 1. Graph of sensitivity (TPR) versus k (Number of nearest neighbours)

Decision Tree (DT):

- The minsplit hyperparameter was tested across different values (from 1 to 30) to determine the optimal number of observations required to split a node.
- The best performance was achieved with minsplit = 17, which yielded a balance between model complexity and sensitivity.

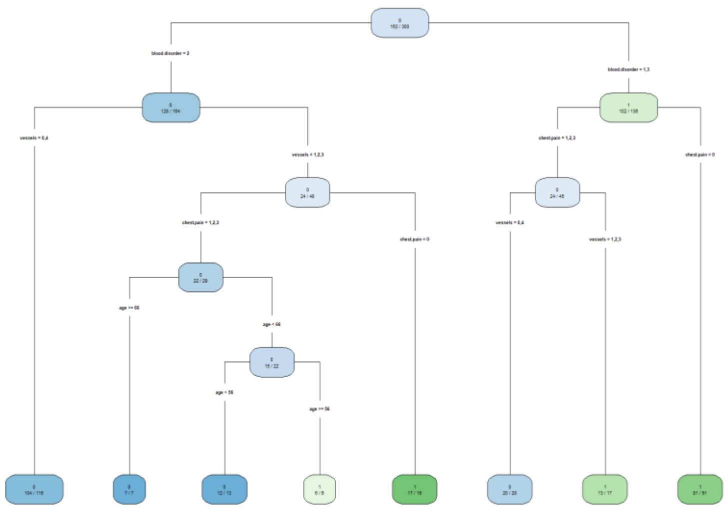

Image 2. Finalized decision tree model with minsplit = 17.

Logistic Regression (LR):

- Features with high p-values and those determined to be insignificant in exploratory data analysis (EDA) were removed (e.g., bp, chol, fbs, rest.ecg, angina, age).
- A simple logistic regression model was then fitted, and TPR was evaluated using cross-validation.
- The final model selected yielded the best TPR without excessive complexity.

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2. PERFORMANCE COMPARISON

After finding the optimal versions of each classifier, we evaluated their performance on the full dataset (300 observations) using the following metrics:

- **True Positive Rate (TPR)**: The proportion of actual positive cases correctly identified.
- **Precision**: The proportion of positive predictions that are correct.
- **ROC Curve**: A plot of TPR versus the False Positive Rate (FPR).
- **Area Under the Curve (AUC)**: The overall performance metric for model discrimination.

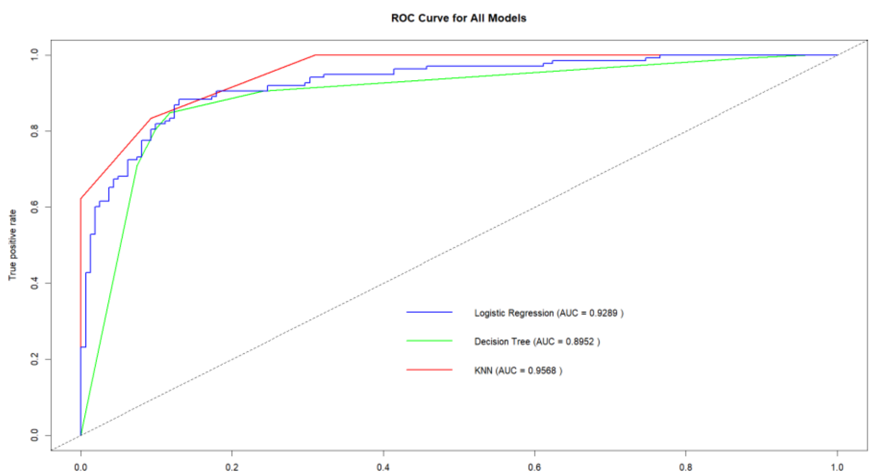

Image. 3 ROC Curve and AUC values for KNN, DT, and LR

Model	AUC	Sensitivity (TPR)	Precision
KNN	0.957	0.833	0.885
Decision Tree (DT)	0.895	0.848	0.860
Logistic Regression	0.929	0.833	0.852

Table 4. Performance comparison based on AUC, Sensitivity (TPR), and Precision

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3. ADVANTAGES AND LIMITATIONS

Classifier	Pros	Cons
KNN	- Highest AUC (~0.957) – excellent overall accuracy - High precision (~0.885) - Captures non-linear patterns well - No model training needed	- Sensitive to scaling and choice of k - Computationally intensive at prediction - Encoded categorical may oversimplify relationships
Decision Tree	- Highest TPR (~0.848) – strong at identifying true positives - Highly interpretable and visualizable - Handles raw and categorical features well	- Lower AUC (~0.895) than KNN and LR - Prone to overfitting if not properly pruned - Sensitive to small changes in data
Logistic Regression	- Balanced performance (AUC ~0.929, TPR ~0.833) - Statistically interpretable coefficients - Efficient and good for feature selection	- Assumes linearity in log-odds - Less flexible for non-linear patterns - Dependent on correct variable selection

Table 5. Pros and cons of the classifiers fitted

CONCLUSION: THE BEST MODEL/CLASSIFIER

After evaluating K-Nearest Neighbours, Decision Tree, and Logistic Regression classifiers using 5-fold cross-validation on the heart disease dataset, Logistic Regression emerges as the most suitable model. While KNN achieved the highest AUC and precision, it is sensitive to feature scaling and less robust with categorical variables. Decision Tree models showed strong sensitivity but are prone to overfitting and offered lower AUC compared to other methods.

Logistic Regression strikes the best balance between predictive performance and model interpretability. With an AUC of approximately 0.929, a sensitivity of ~0.833, and high precision, it demonstrates strong and consistent performance. Additionally, its simplicity and transparency make it particularly well-suited for medical applications, where understanding the influence of predictors is crucial.

Therefore, Logistic Regression is recommended as the optimal classifier for this dataset.